

COMPUTATIONAL GENOMICS FINAL PROJECT

# Who left this DNA?

A mystery Nanopore fecal sample, six possible host animals, one microbial co-star, and a small pile of evidence.

## Group D

Host origin, microbiome, assembly, and validation from curated project outputs.

# We followed the final assignment step by step

1

## QC/A

Describe FASTQ statistics, visualize metrics, and filter low-quality reads.

2

## Comparative alignment

Align reads to the six candidate animal genomes and choose the best match.

3

## Functional region

Find one sequenced region with a known or predicted function.

4

## Microbiome exploration

Classify remaining reads and visualize top taxa.

5

## Species assembly

Assemble one abundant microbe and compare it to a reference genome.

The README is the presentation structure: each section below is one assignment step.

GET YOUR DATA

# Starting point: Team D Nanopore reads and run reports

## Input

Long-read FASTQ data from the Group D sample.

## Reports

MinKNOW, NanoComp, and NanoPlot summaries from the sequencing run.

## Question

Which animal feces sample was sequenced, and what microbes are present?

## Team D

our sample group

## ONT

Oxford Nanopore long reads

## 6

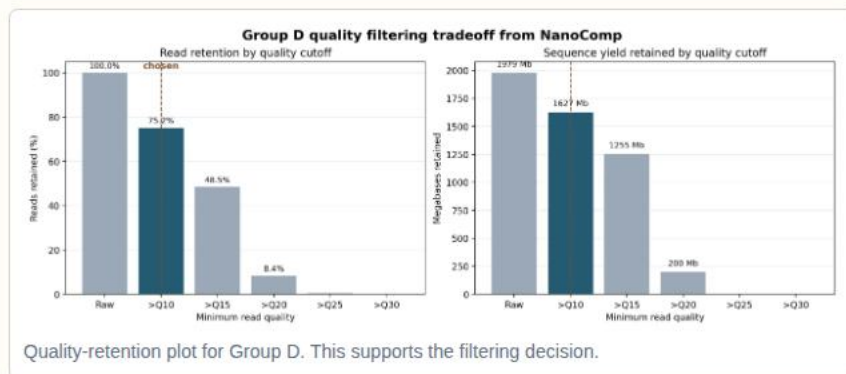
candidate animal genomes

## 1

abundant microbe selected for assembly

## STEP 1: QC/A

# Describe the sequences and decide how to filter



### Filtering logic:

Keep enough data for downstream analysis while removing the weakest reads.

### Why NanoPlot/NanoComp?

The README asks for FASTQ statistics and visualizations. NanoPlot and NanoComp are better suited than FastQC for Nanopore read length and quality summaries.

## STEP 1: QC/A

# The visual reports stayed part of the analysis



This answers the README request for visual metrics and explains why the filtered reads were usable.

STEP 2: COMPARATIVE ALIGNMENT ASSESSMENT

## Six candidate animal genomes



**House mouse**

Good cameo, weak evidence.



**Kangaroo**

Memorable, but not our sample.



**Domestic cat**

Quietly sitting on the strongest signal.



**Norway rat**

Similar enough to test, not enough to win.



**Golden jackal**

Second place in some counts.



**Little brown bat**

Useful negative control energy.

The README asks us to decide which animal feces sample we sequenced by comparing alignments to these genomes.

STEP 2: COMPARATIVE ALIGNMENT ASSESSMENT

## Our method for deciding which alignment is better

### Mapped reads

How many reads aligned to each candidate genome?

### Aligned bases

How much total sequence was aligned?

### MAPQ

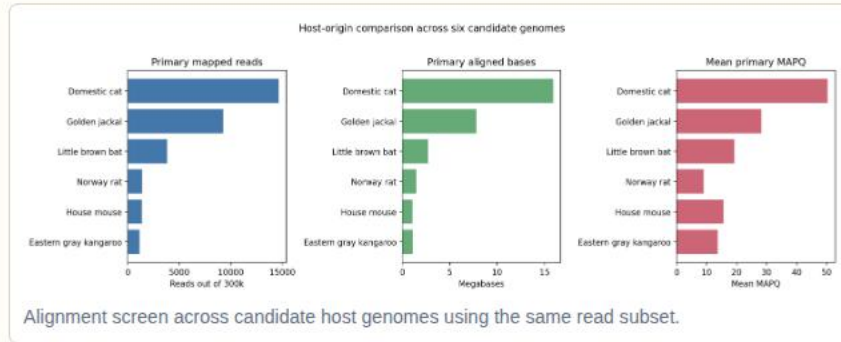
A confidence score from the aligner. Higher is better.

### Tool choice:

The README lists minimap2 as a useful aligner for long noisy DNA reads. We used minimap2 instead of BWA because this is Nanopore data, not short paired-end Illumina data.

## STEP 2: COMPARATIVE ALIGNMENT ASSESSMENT

# The domestic cat genome had the strongest signal



### Chosen animal: domestic cat

It had the most primary mapped reads and the most aligned bases among the candidate genomes.

### Expected low alignment rate:

The README warns that fecal samples contain many microbiome reads, so low host alignment is expected.



## STEP 2: COMPARATIVE ALIGNMENT ASSESSMENT

# Cat wins by count and confidence

| Candidate              | Primary mapped | Aligned bases | Mean MAPQ |
|------------------------|----------------|---------------|-----------|
| 1 Domestic cat         | 14,669         | 15.9 Mbp      | 50.3      |
| 2 Golden jackal        | 9,251          | 7.8 Mbp       | 28.3      |
| 3 Little brown bat     | 3,818          | 2.7 Mbp       | 19.2      |
| Mouse / rat / kangaroo | 1,155-1,393    | 1.1-1.5 Mbp   | 9.0-15.6  |

### Why not jackal?

Golden jackal had the next strongest signal, but fewer mapped reads, fewer aligned bases, and much lower MAPQ.

### Final host call:

Domestic cat.

STEP 2 RESULT

# The sample origin is domestic cat

The README asks us to present the chosen animal and explain how we decided.  
This is the answer.

Top mapped reads

Top aligned bases

Top MAPQ



### STEP 3: FUNCTIONAL REGION THAT WAS SEQUENCED

## After choosing cat, we looked for a functional region

#### README instruction:

Locate one genome region with functional capabilities, investigate its role, and present it in class.

#### What we did:

Used the cat genome annotation file and counted which annotated gene regions overlapped our cat-aligned reads.

1

#### Cat-aligned reads

Start only after the host call.

2

#### Cat annotation file

A table of gene coordinates from NCBI.

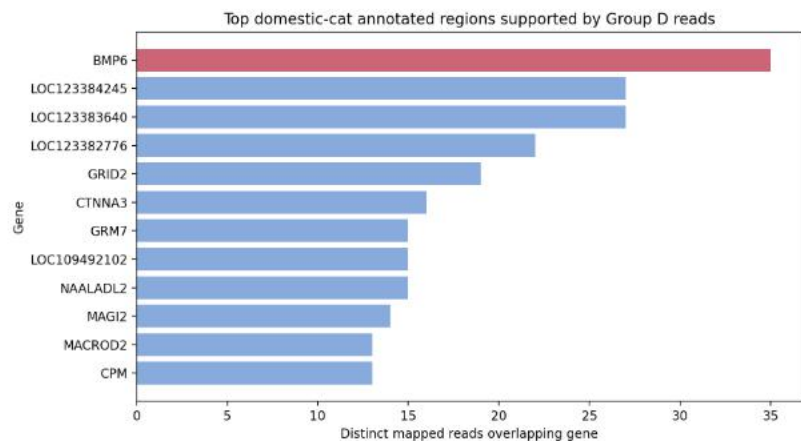
3

#### Overlap count

Which labeled region received the most read support?

### STEP 3 RESULT

## BMP6 was our functional-region example



BMP6 had the highest read count among annotated cat gene overlaps in our summary.

#### Why BMP6?

It was the top labeled gene region in the overlap table: 35 reads overlapped the BMP6 interval.

#### What BMP6 is:

BMP6 is a mammalian gene in a signaling family often discussed in development, bone/cartilage biology, and iron regulation.

#### What we are not saying:

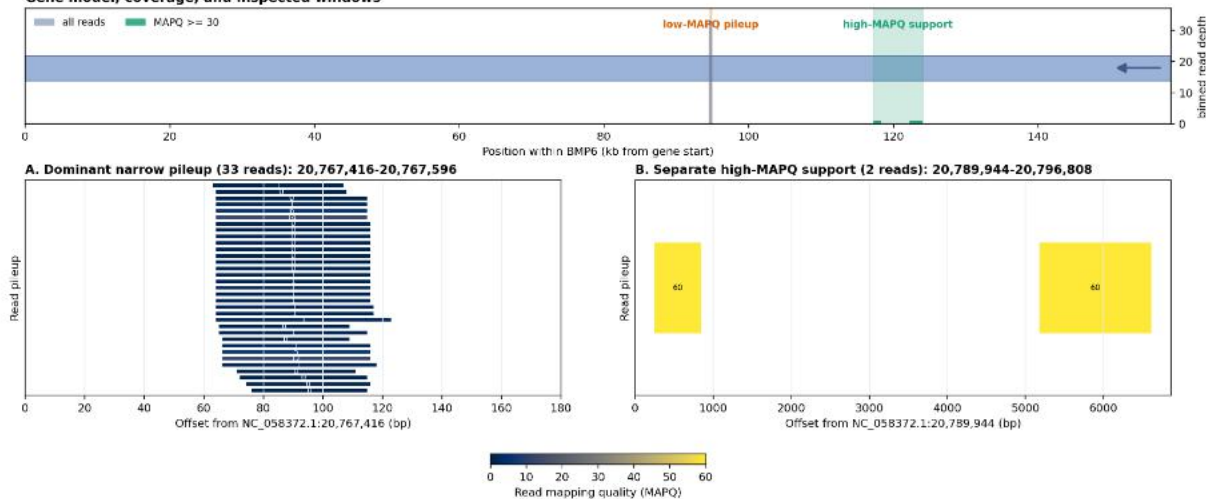
We are not saying this cat had a BMP6 mutation, phenotype, or special biology. We only found that this annotated region was represented in the host reads.

# Genome-browser style view of the BMP6 region

## BMP6 host-region evidence: genome-browser-style view

NC\_058372.1:20,672,770-20,831,131 | 158.4 kb gene | 35 overlapping reads | 2 reads with MAPQ  $\geq 30$

### Gene model, coverage, and inspected windows



IGV-style BMP6 region panel: genomic coordinates, read pileup, and annotation track.

### Why include this view?

Summary tables can hide awkward alignments. A genome-browser view lets us sanity-check whether the reads really sit near the annotated region.

### Read it carefully:

The signal is concentrated and should be treated as supporting evidence, not as a functional discovery.

#### STEP 4: MICROBIOME EXPLORATION

## Classify the remaining reads

### After host screening: who else is in there?

The README says most reads in fecal samples come from microbes living alongside the animal.

#### Why Kraken2?

The README lists Kraken2 and Centrifuge as taxonomy-classification tools. We used Kraken2 because it is fast, works with Nanopore reads, and has small downloadable databases.

#### Output:

Tables and plots of top genera/species in the microbiome profile.

#### STEP 4 METHOD

# Kraken2 uses k-mers for taxonomy classification

1

## Split reads

Kraken2 breaks reads into short k-mers.

2

## Compare to database

The database connects k-mers to known organisms.

3

## Summarize taxa

The result is a microbiome classification table.

### Why this fits the task:

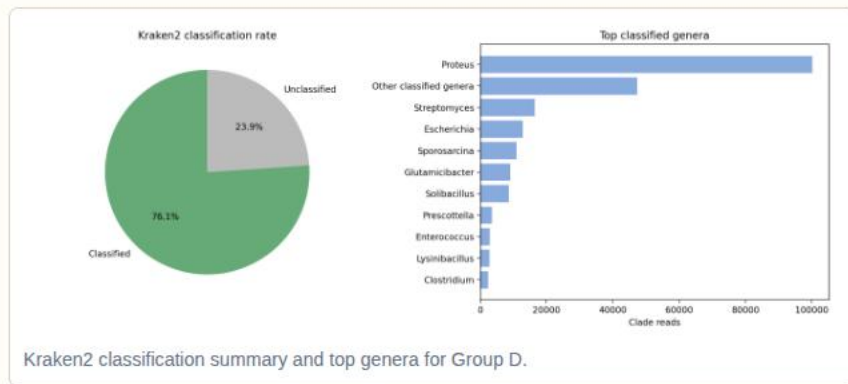
The assignment asks for microbiome composition, diversity, and top taxa. Classification is exactly that question.

### Why we still keep caveats:

The answer depends on the database. Closely related organisms can share many k-mers.

#### STEP 4 RESULT

## *Proteus* dominates the genus-level profile



**35.11%**

Proteus at genus level

**5.77%**

Streptomyces

**4.49%**

Escherichia

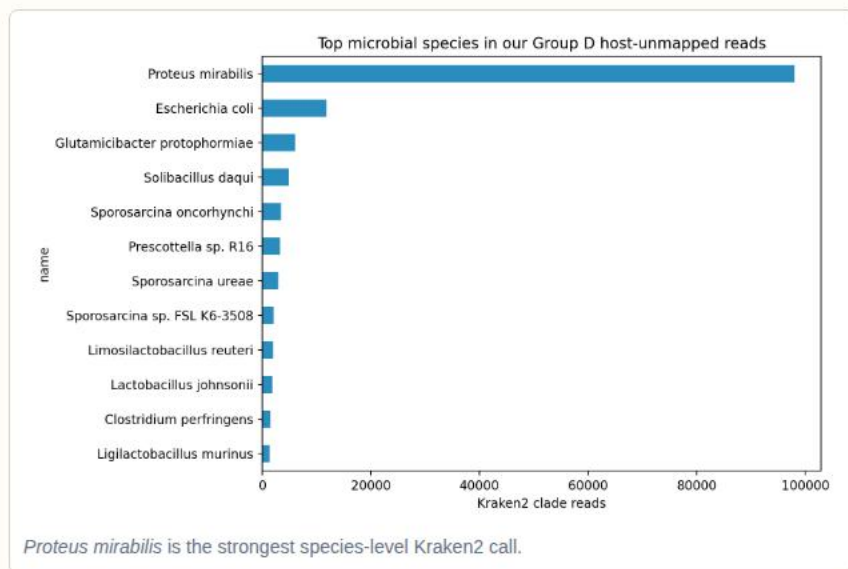
**3.84%**

Sporosarcina



#### STEP 4 RESULT

## The dominant species is *Proteus mirabilis*



#### Top species call:

*Proteus mirabilis*, 34.35% of reads in the Kraken2 report.

#### The biological story so far:

A domestic cat fecal sample with a strong *Proteus mirabilis* microbial signal.

## STEP 5: SPECIES ASSEMBLY

# Choose one abundant microbe and assemble its genome

1

### Pick species

*Proteus mirabilis* was the top species-level call.

2

### Assemble reads

Flye assembled the selected long reads into contigs.

3

### Assess correctness

dnadiff compared the assembly to a reference genome.

### What Flye did:

Flye is the assembler: it reconstructs contigs from noisy long reads. It is not the classifier and not the validation step.

### Why Flye?

Flye is built for noisy long reads and is commonly used for bacterial Nanopore assemblies. Canu is powerful but heavier, Raven is fast and lightweight, and Shasta is excellent in some long-read contexts. Flye was a balanced choice for this final-project dataset.

STEP 5 RESULT

## Assembly statistics

**README metric:**

Report how many contigs there are and the length of the longest one.

**Interpretation:**

A fragmented assembly is expected from a mixed fecal sample.

**3.86 Mbp**

assembly size

**24**

contigs

**577,987 bp**

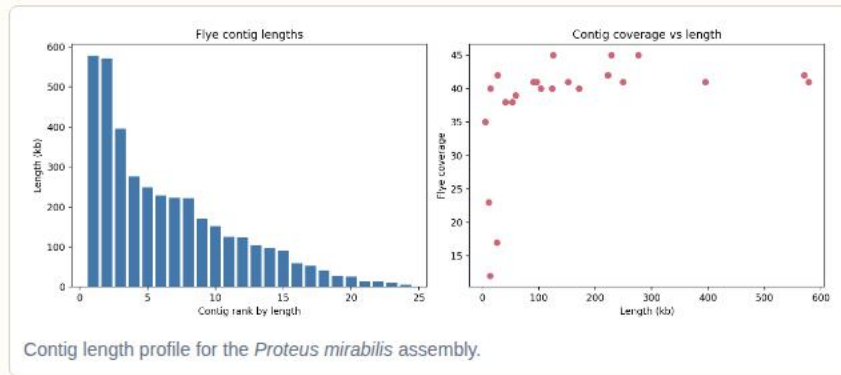
longest contig

**249,377 bp**

N50

#### STEP 5 RESULT

## The assembly is fragmented, but biologically useful



#### What this means:

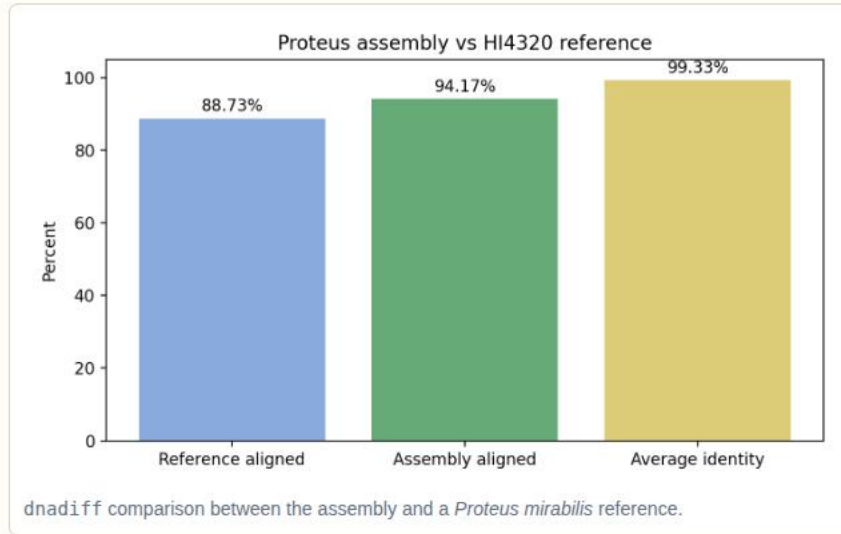
A perfect bacterial assembly would be closer to one complete chromosome. Here we got a partial but coherent assembly from a mixed fecal sample.

#### Good enough for:

Checking whether the dominant microbe can be reconstructed and compared against a reference.

## STEP 5 VALIDATION

# Compare the assembly to a reference genome



### Average identity: 99.33%

The assembled sequence aligns very closely to the reference where it aligns.

### Coverage is not complete.

88.73% of the reference and 94.17% of the assembly aligned in the comparison.

### Why dnadiff?

The README suggests dnadiff from MUMmer for comparing the assembly to a reference genome. We used it to report identity and aligned coverage.

## SUMMARY

# Main findings, in README order

**Animal origin:**

Domestic cat.

**Functional region example:**

BMP6 gene region, chosen because it had the strongest annotated host-read overlap.

**Dominant microbe:**

*Proteus mirabilis*.

**Assembly:**

3.86 Mbp, 24 contigs, N50 249,377 bp, 99.33% identity.

# The cat did it.

*Proteus mirabilis* left the microbial fingerprints.

**Caveat 1**

BMP6 is a functional-region example, not a phenotype claim.

**Caveat 2**

Kraken2 classifications depend on the database.

**Caveat 3**

The assembly is not a finished genome.

**Full report**

Curated GitHub Pages report

**Notebook**

Rendered reproducible analysis

**PDF deck**

Downloadable presentation